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SIMCU-STATE: Smart Urban Freight Monitoring System for State-Level

Logistics Planning:

Analysis of Inter-Regional Freight Flow Impacts on Urban Node Safety and

Traffic Performance

within the São Paulo State Logistics and Infrastructure Plan (PLI-SP 2050)

Framework

Using Integrated Multimodal ITS Architecture and SEMIL State Datasets

Dissertation Proposal for PhD Program

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TABLE OF CONTENTS

Abstract	3
Research Overview	3
Problem Statement Summary	3
Methodological Approach	4
Expected Contributions	4
1. Introduction and Research Context	5
1.1 Background on Urban Freight Transport in São Paulo State	5
1.2 The PLI-SP 2050 Framework and the Urban Node Challenge	7
1.3 Intelligent Transport Systems as Enabling Technology	10
2. Comprehensive Literature Review	13
2.1 Urban Freight Transport Impact Assessment	13
2.2 Road Safety Dimensions of Inter-Regional Freight at Urban Nodes	15
2.3 Traffic Performance Analysis at Freight Corridors	18
2.4 Intelligent Transport Systems for State-Scale Freight Monitoring	19
2.5 Data-Driven Safety and Performance Modeling	21
3. Research Problem Statement	23
3.1 Core Problem Definition	23
3.2 Specific Challenges in State-Level Freight-Urban Integration	25

3.3 Research Questions	26
4. Research Objectives	28
4.1 Primary Objective	28
4.2 Specific Objectives	28
5. Detailed Research Methodology	32
5.1 Phase 1: Foundation and Data Preparation	32
5.2 Phase 2: SIMCU-State System Development	33
5.3 Phase 3: Safety Impact Analysis at Urban Nodes	34
5.4 Phase 4: Traffic Performance Analysis	34
5.5 Phase 5: System Validation and Evaluation	35
5.6 Phase 6: Policy Formulation and Dissemination	35
6. Data Sources: PLI-SP 2050 and SEMIL Datasets	37
6.1 Rationale for State-Level Data Integration	37
6.2 SEMIL DataGEO and Origin-Destination Matrix Deployment	39
6.3 Supporting State Datasets and Academic Literature	40
7. Expected Contributions and Significance	42
7.1 Theoretical Contributions	42
7.2 Practical Contributions	42

7.3 Policy Implications	43
References	44

ABSTRACT

São Paulo State's inter-regional freight network generates substantial yet poorly quantified safety, congestion, and equity impacts at urban nodes the cities and logistics hubs where state-level freight corridors intersect with local traffic systems. These impacts are structurally unequal: Cardoso (2025) documents how freight infrastructure investment and its externalities are distributed asymmetrically across São Paulo's centro-periphery axis, with peripheral municipalities bearing disproportionate traffic mortality and congestion costs relative to their infrastructure quality. State-level planning instruments, principally the PLI-SP 2050 framework and SEMIL's 62 multimodal origin-destination matrices, characterise freight flows at macro scale and embed PLI-SP 2050's Socioeconomic Characterisation data for all São Paulo State planning zones, while urban traffic management remains confined to municipal sensor networks and retrospective crash data (Marchetti & Wanke, 2021; SEMIL, 2026). No validated analytical framework exists to bridge these scales translating PLI-SP 2050's macro socioeconomic and multimodal intelligence into localised urban node impact assessments in real time and the absence of this bridge has concrete consequences for the equity and sustainability of São Paulo State's logistics governance.

This dissertation proposes SIMCU-State (Smart Urban Freight Monitoring System State Integration Module), an Intelligent Transport System (ITS) architecture that operationalises PLI-SP 2050 OD matrix data to monitor and predict road safety risks and traffic performance degradation at São Paulo State urban nodes. The research employs a six-phase mixed-methods design across a 48-month PhD timeline: OD matrix disaggregation using doubly-constrained entropy maximisation following Wang and Holguin-Veras (2008); SIMCU-State architecture development with DataGEO API integration benchmarked against ISO 21217 and IEEE 2413

standards; Poisson and negative binomial safety modelling calibrated with state freight exposure metrics (Lord & Mannering, 2010); traffic performance assessment through level-of-service and travel time reliability indices; quasi-experimental difference-in-differences validation across PLI-SP 2050 node typologies; and evidence-based policy formulation through structured stakeholder engagement.

The dissertation contributes a validated state-to-urban freight impact translation framework; a multi-institutional ITS deployment architecture extending beyond city-scale precedents; and empirical characterisation of safety and performance patterns across São Paulo State's logistics corridor typologies. Practical outputs include deployable DataGEO-integrated monitoring tools, validated disaggregation procedures directly compatible with SEMIL's PLI-SP 2050 planning workflows, and replication guidelines scalable to the state's full urban node inventory. Collectively these contributions transform SIMCU-State from a single-site study into foundational analytical infrastructure for São Paulo State freight governance.

1. INTRODUCTION

1.1 Background on Urban Freight Transport in São Paulo State

1.1.1 Economic Significance and Growth Trends

São Paulo State constitutes the most economically productive territorial unit in Latin America, generating approximately 32% of Brazilian GDP and sustaining an industrial base whose logistics requirements shape the configuration of the national freight network (IBGE, 2023). The state's freight system is not merely an internal distribution challenge but a macro-economic infrastructure whose performance determines supply chain costs across the entire Brazilian market. Urban nodes the cities and metropolitan agglomerations where inter-regional freight corridors intersect with local distribution systems represent the critical stress points in this network, absorbing the consequences of state-level logistics decisions in the form of congestion externalities, road safety incidents, and infrastructure degradation.

The socioeconomic context that structures these freight impacts constitutes the foundational macro layer upon which SIMCU-State's analytical framework is built. São Paulo State's development trajectory has produced one of the most spatially unequal urban systems in Latin America, characterised by a pronounced centro-periphery divide documented in PLI-SP 2050's own Socioeconomic Characterisation component: formal employment, logistics infrastructure, and transport investment have historically concentrated in an expanded metropolitan centre, while peripheral municipalities along the state's primary freight corridors SP-310, SP-330, SP-348, and SP-60 host both the highest inter-regional freight volumes and the most acute deficits in road safety infrastructure, traffic management capacity, and vulnerable road user protection (SEMIL, 2026). Research by Cardoso (2025) on logistics and inequality in the São Paulo Metropolitan Region documents how freight network design systematically disadvantages peripheral

populations, creating a spatial coincidence between logistics intensity and socioeconomic vulnerability that conventional freight planning instruments do not operationalise. The Ecological-Economic Zoning of São Paulo State (ZEE-SP), integrated within the DataGEO platform, documents how logistics corridor development intersects with zones of high ecological sensitivity and socioeconomic fragility, reinforcing the imperative to account for distributional externalities in freight planning decisions (SEMIL, 2026). This macro socioeconomic characterisation connects SIMCU-State directly to Sustainable Development Goal 11 (Sustainable Cities and Communities) and SDG 9 (Industry, Innovation, and Infrastructure), both of which require that infrastructure investment be evaluated not only for aggregate efficiency gains but for equitable safety and accessibility outcomes across socioeconomic strata.

The economic significance of this challenge is underscored by the scale of state freight activity. São Paulo State manages a multimodal logistics network encompassing road, rail, waterway, and pipeline corridors whose combined freight volumes generate externalities of national consequence. The PLI-SP 2050 initiative, developed by the Secretariat of Environment, Infrastructure, and Logistics (SEMIL), represents the state government's strategic response to these systemic pressures, establishing a long-horizon planning framework aimed at reducing logistics costs, improving multimodal integration, and enhancing the environmental sustainability of freight operations across the state's territory (SEMIL, 2026). The research value of this initiative for transportation engineering lies in the unprecedented data infrastructure it has assembled: 62 origin-destination matrices covering the state's full multimodal freight network, centralized within the DataGEO geospatial platform.

This dissertation situates itself at the productive intersection of this state planning initiative and the technical literature on urban freight impact assessment. The core argument is that SEMIL's

state-level data infrastructure creates a historically unique opportunity to construct validated analytical frameworks for translating macro-logistics planning decisions into micro-urban impact predictions a translation that has remained methodologically underdeveloped despite its operational significance. The e-commerce intensification, last-mile proliferation, and just-in-time supply chain pressures that are reshaping global urban freight (Ferreira et al., 2023; Dutra et al., 2021) are simultaneously reshaping the state's urban nodes, demanding analytical frameworks that can operate across planning scales.

This dissertation addresses a specific and consequential problem: São Paulo State's logistics governance system generates detailed macro-scale freight demand intelligence through PLI-SP 2050 yet lacks the analytical bridge to translate that intelligence into urban-level safety and congestion impact predictions. The consequence is that infrastructure investment decisions affecting tens of millions of urban residents are made without systematic knowledge of their localised traffic consequences, exposing the state's urban populations to preventable freight-related crash risk and congestion costs that go unmeasured, unmodelled, and therefore unmitigated. Brazil's logistics sector accounts for approximately 11% of GDP (CNT, 2022), and São Paulo State bears a disproportionate share of both the economic opportunity and the externality burden this system generates. SIMCU-State is designed to close this governance gap by constructing the first validated state-to-urban freight impact translation framework in the Brazilian context.

1.1.2 Unique Challenges in São Paulo State's Urban Environments

The urban environments of São Paulo State present distinctive freight management challenges that reflect the state's extraordinary internal diversity. The Metropolitan Region of São Paulo, with its concentrated industrial base and approximately 22 million inhabitants (IBGE, 2022), generates and absorbs freight volumes that dwarf those of most national capitals. Interior cities such as

Campinas, Ribeirão Preto, São José dos Rio Preto, and São Carlos occupy secondary nodes in the state logistics hierarchy, serving simultaneously as regional distribution hubs and as intermediate points on inter-regional freight corridors (Vieira et al., 2016).

Mixed traffic conditions prevail across the state's urban fabric, with freight vehicles sharing road infrastructure with high motorcycle volumes, substantial pedestrian activity, and expanding cycling infrastructure in circumstances where separation standards are often inadequate. Regulatory enforcement capacity is geographically uneven, creating spatial variation in compliance with weight restrictions, time-of-day delivery rules, and loading zone regulations that complicates standardized analysis. The age heterogeneity of the freight vehicle fleet ranging from modern logistics operators employing advanced telematics to informal owner-operators maintaining aging vehicles without electronic safety systems generates corresponding heterogeneity in safety and emissions profiles that aggregate state-level planning instruments struggle to capture (CNT, 2022).

The PLI-SP 2050 framework explicitly acknowledges this complexity, positioning urban node analysis as a distinct analytical challenge requiring specialized methodological treatment. The state's investment in the DataGEO platform and the assembly of 62 OD matrices reflects a recognition that effective state logistics planning requires understanding the urban consequences of inter-regional freight decisions a translation challenge that remains methodologically underdeveloped despite its operational significance.

1.1.3 Policy and Regulatory Landscape

The institutional context for this research is defined by the convergence of state strategic planning ambitions with federal mobility policy frameworks. The National Policy for Urban Mobility (Law 12.587/2012) established sustainable transport planning principles that have been progressively

operationalized through state-level instruments in São Paulo, most notably through SEMIL's integrated infrastructure and environment mandate. The PLI-SP 2050 initiative extends this institutional evolution to the logistics domain, establishing a planning horizon and data infrastructure that create favorable conditions for academically rigorous, policy-relevant research.

The São Paulo State Climate Adaptation and Resilience Plan (PEARC), the State Climate Action Plan (PAC 2050), and the State Energy Plan 2050 (PEE 2050) contextualise the freight sustainability dimensions of this research within the state's integrated environmental and energy governance architecture (SEMIL, 2026). PEARC establishes climate resilience requirements for transport infrastructure under projected precipitation intensification and heat stress scenarios across São Paulo State, requirements that freight corridor management must satisfy if PLI-SP 2050 investments are to remain functional through the 2050 planning horizon. PAC 2050 sets sectoral emissions reduction trajectories requiring measurable decarbonisation of road freight a target whose urban node implications SIMCU-State is positioned to quantify through real-time monitoring of freight vehicle emissions and exposure patterns. PEE 2050 defines the energy transition pathway for São Paulo State's commercial vehicle fleet, projecting progressive electrification scenarios that will alter fleet composition, maintenance profiles, and infrastructure requirements at the urban nodes SIMCU-State monitors. These three instruments collectively define the sustainability envelope within which PLI-SP 2050's freight governance must operate, and SIMCU-State's architecture is designed to supply the empirical monitoring evidence needed to track compliance with their requirements at the urban scale.

1.2 The PLI-SP 2050 Framework and the Urban Node Challenge

1.2.1 The PLI-SP 2050 Initiative: Scope and Strategic Objectives

The São Paulo State Logistics and Infrastructure Plan 2050 constitute the most comprehensive state-level freight planning instrument in Brazilian history, integrating multimodal network analysis, socioeconomic characterization, climate scenario planning, and institutional capacity development within a unified strategic framework. Developed through SEMIL in collaboration with federal agencies, research institutions, and private sector stakeholders, PLI-SP 2050 addresses the structural logistics inefficiencies that have historically elevated costs and constrained the state's economic competitiveness. The plan's analytical foundation rests on a suite of geospatial and quantitative tools assembled within the DataGEO platform, including the 62 origin-destination matrices that characterize freight flows across the state's multimodal network for the base planning year (SEMIL, 2026).

The strategic objectives of PLI-SP 2050 encompass logistics cost reduction through multimodal diversification, infrastructure investment prioritization based on system-level performance modeling, environmental impact mitigation aligned with PAC 2050 targets, and institutional coordination improvements across the state's fragmented transport governance architecture. These objectives are analytically ambitious, requiring simulation capabilities that can model the consequences of infrastructure investments, regulatory changes, and modal shift incentives across spatial scales ranging from state-wide network performance to corridor-level traffic flow. This multi-scale analytical requirement creates the precise methodological niche that SIMCU-State is designed to fill: the translation of state-level freight demand modeling into urban node impact assessment.

1.2.2 Urban Nodes in State Logistics Systems: Definition and Classification

Urban nodes in the PLI-SP 2050 analytical framework are defined as the municipalities and metropolitan agglomerations where inter-regional freight flows generate measurable impacts on urban traffic systems, road safety records, and local infrastructure condition. This definition encompasses a spectrum from the São Paulo Metropolitan Region, which functions as the state's primary logistics hub and the convergence point for flows from all state corridors, to secondary cities along major state highways such as SP-310, SP-330, and SP-348 that serve as intermediate distribution and transshipment points (Guerin et al., 2021).

The classification of urban nodes by their functional role in the state logistics network primary hubs, secondary distribution centers, corridor waypoints, and agricultural freight gateways is analytically important because each node type generates a distinctive pattern of freight-urban interaction. Primary hubs experience the highest absolute freight volumes and the most complex multimodal interactions, but may also possess more developed traffic management infrastructure. Secondary cities and corridor waypoints often face disproportionate freight impacts relative to their infrastructure capacity, particularly where state highways traverse urban areas without adequate grade separation or freight bypass routing.

The socioeconomic characterisation component of PLI-SP 2050 provides demographic, economic, and land use data by planning zone that is essential for contextualising freight impact assessments across this urban node hierarchy (SEMIL, 2026). This characterisation is not merely background context but the macro analytical layer that enables SIMCU-State's safety and performance models to account for the structural inequality conditions that determine how freight impacts are distributed. The logistics decentralisation process documented by Guerin et al. (2021) whereby distribution centre operations have migrated from the inner industrial zone to peripheral locations

along the SP-310, SP-330, and SP-348 corridors has created a spatial coincidence between high inter-regional freight exposure and low infrastructure capacity at precisely the peripheral nodes where PLI-SP 2050's corridor typology identifies secondary distribution centres and corridor waypoints. The ZEE-SP ecological-economic zoning layers within DataGEO further differentiate node contexts by documenting land use transitions, environmental sensitivity, and socioeconomic fragility at the meso-scale, enabling SIMCU-State's disaggregation procedures to incorporate land use heterogeneity as a structural predictor rather than treating it as a uniform covariate. Integration of this multi-layered characterisation data with traffic safety records and performance monitoring enables the differentiated analysis of freight impacts across node typologies, essential for evidence-based policy targeting aligned with PLI-SP 2050's logistics objectives and São Paulo State's SDG 11 commitments.

1.2.3 The Analytical Gap: State-Level Planning Meets Urban Impact Reality

The fundamental analytical challenge motivating this dissertation is the mismatch between the spatial and temporal scales at which PLI-SP 2050 operates and the scales at which urban freight impacts manifest. State-level OD matrices characterize inter-zonal freight flows in terms of origin, destination, commodity type, and transport mode, but do not directly describe the traffic stream compositions, loading/unloading patterns, and vehicle routing behaviors that determine safety risks and performance degradation at specific urban locations. Bridging this gap requires a methodological framework that can disaggregate state-level demand estimates into urban network loading profiles, calibrate these profiles against observed traffic and safety data, and use the resulting models to generate actionable urban-level predictions.

This analytical gap is particularly acute because state freight planning decisions infrastructure investment priorities, regulatory frameworks, modal incentive structures directly determine the

volume and composition of freight flows arriving at urban nodes, yet the urban consequences of these decisions are not systematically modeled within PLI-SP 2050's current analytical toolkit. The research community that has examined São Paulo State freight logistics has predominantly focused on supply chain economics and network-level performance (Guerin et al., 2021; Vieira et al., 2016) without developing the urban impact assessment methodologies that would enable PLI-SP 2050 to fully account for externality costs in its prioritization framework.

1.3 Intelligent Transport Systems as Enabling Technology

1.3.1 Evolution from Static to Real-Time Monitoring Architectures

Intelligent Transport Systems have undergone substantial architectural evolution over three decades, progressing from isolated automated applications toward integrated, network-scale monitoring architectures capable of processing heterogeneous real-time data streams. Early ITS deployments were characteristically reactive and single-function: traffic signal control, incident detection, and electronic toll collection operated as independent systems without systematic data integration or cross-function analytical capability. Contemporary architectures, exemplified by the InterSCity platform developed for São Paulo State smart city applications, emphasize real-time data fusion across multiple sensor modalities, cloud and edge computing integration, and open API structures that enable third-party analytical applications (Institution of Engineering and Technology, 2023).

The relevance of this architectural evolution to PLI-SP 2050 integration lies in the interoperability requirements it establishes. Connecting SIMCU-State's urban monitoring capabilities to SEMIL's DataGEO platform requires standardized data exchange protocols, compatible spatial referencing systems, and temporal synchronization mechanisms that allow state-level OD matrix updates to propagate through the urban impact modeling pipeline in near real-time. The platform-layer

architecture of modern ITS systems, employing microservice designs with standardized APIs, creates the technical foundation for this integration.

1.3.2 IoT, Computer Vision, and State Database Integration

SIMCU-State's monitoring architecture leverages three complementary technology categories alongside the state data integration layer that distinguishes it from conventional city-scale ITS deployments. IoT sensors encompassing inductive loops, radar, lidar, and acoustic detectors provide real-time vehicle detection and classification data at urban node entry and exit points, enabling continuous estimation of freight vehicle exposure metrics. Modern sensors can distinguish vehicle categories relevant for state freight analysis, including light commercial vehicles, medium trucks, heavy trucks, and articulated combinations, with classification accuracy sufficient for safety and performance modeling applications (Institution of Engineering and Technology, 2023).

Computer vision from traffic cameras enables behavioral observation extending beyond vehicle detection to include loading and unloading activity identification, double-parking detection, and intersection conflict analysis. The integration of computer vision outputs with state OD matrix data enables a novel analytical capability: the real-time calibration of state-level flow estimates against observed urban node traffic compositions, providing a continuous feedback mechanism that improves the accuracy of PLI-SP 2050's urban impact predictions over time. This calibration function is analytically important because OD matrices derived from state-level surveys and simulation models inevitably incorporate estimation errors that accumulate in urban impact projections without field-based correction.

State database integration connecting SIMCU-State to DataGEO geospatial layers, INFOSIGA-SP crash records, DETRAN-SP vehicle registration data, and PLI-SP 2050 socioeconomic

characterization files constitutes the most distinctive technical component of the proposed architecture. The DataGEO platform, which centralizes São Paulo State's geospatial information infrastructure across environmental, infrastructure, and logistics domains, provides both the base data layers necessary for spatial analysis and the API connectivity required for real-time data exchange (SEMIL, 2026).

1.3.3 Applications in State-Scale Freight Traffic Management

The state-scale application context for SIMCU-State differs from conventional city-scale ITS freight deployments in analytically significant ways. While city-scale systems typically address local delivery management loading zone optimization, time-of-day restrictions, last-mile routing state-scale applications must address the consequences of inter-regional transport decisions on urban node performance. This includes the traffic and safety implications of freight modal shift incentives, infrastructure investment prioritization, and logistics corridor regulation the core policy instruments of PLI-SP 2050.

Dynamic Access Control (DAC) systems, operationalised in European smart corridor programmes such as FENIX, regulate vehicle access based on real-time traffic conditions, emissions parameters, and infrastructure capacity constraints, offering adaptive management capabilities that static Urban Vehicle Access Regulations (UVARs) cannot provide (European Commission, 2022). The integration of DAC principles with PLI-SP 2050's state-level planning framework creates a hierarchical management architecture spanning from state corridor routing guidance to urban node access control a management hierarchy that SIMCU-State is designed to technically support.

SIMCU-State extends conventional freight ITS applications by positioning urban node monitoring as a functional module within PLI-SP 2050's planning architecture. Rather than treating urban freight management as a separate functional domain, the system analyzes how inter-regional

freight flows arriving from PLI-SP 2050 network corridors generate safety and performance impacts at urban nodes, enabling trade-off analysis across planning scales that has not previously been systematically available to São Paulo State freight planners.

2. LITERATURE REVIEW

2.0 Theoretical Framework

The literature review is organised within a macro-to-micro analytical logic whose three-layer architecture mirrors the governance structure of PLI-SP 2050 itself. The outermost layer the socioeconomic characterisation of São Paulo State provides the structural inequality context without which freight impact assessments produce technically accurate but socially incomplete predictions. The middle layer the multimodal OD matrix framework translates this socioeconomic geography into quantified freight demand flows across the state's multimodal network. The innermost layer the urban module operationalises those flows as localised safety and performance predictions at specific urban nodes. This layered architecture is theoretically grounded in Urban Logistics Theory (Taniguchi et al., 2014), which conceptualises the city as a logistical system in which freight demand generated by economic activity must be accommodated by transport infrastructure, producing externalities whose distribution across stakeholders creates policy intervention points. The Multi-Level Governance framework (Cidell, 2011) recognises that freight decisions are made at federal, state, metropolitan, and municipal scales, with consequential misalignments between where demand is generated and where impacts manifest. Crucially, the work of Cardoso (2025) on logistics and inequality in the São Paulo Metropolitan Region anchors both frameworks in São Paulo's specific socioeconomic reality: freight network design reproduces and amplifies São Paulo's centro-periphery inequality through selective logistics infrastructure location, asymmetric externality distribution, and the differential political capacity of communities to claim mitigation resources. This equity-centred reading of logistics governance motivates SIMCU-State's explicit commitment to distributional analysis alongside aggregate performance assessment, and its alignment with SDG 9 (resilient and sustainable infrastructure with equitable

access) and SDG 11 (safe, inclusive, and sustainable cities). The review synthesises evidence across five thematic domains to establish the empirical and methodological basis for SIMCU-State's research design, identifying the specific knowledge gaps that each component of the research addresses.

2.1 Urban Freight Transport Impact Assessment

2.1.1 Environmental Externalities: Emissions, Noise, and Air Quality

The environmental externalities generated by urban freight vehicles have been systematically documented across Brazilian and international contexts, establishing the quantitative baseline against which PLI-SP 2050 sustainability targets must be evaluated. A comprehensive life cycle sustainability assessment of light commercial vehicles the dominant last-mile delivery mode revealed that this category achieves the highest aggregate environmental impact score across all assessed categories, with a single-score of 48 kPt. The human carcinogenic toxicity contribution of 4.35 kPt and non-carcinogenic toxicity of 7 kPt document significant public health dimensions that extend beyond conventional carbon accounting (Oliveira et al., 2017).

Heavy-duty diesel freight vehicles contribute disproportionately to urban air pollution relative to their fleet share, emitting substantially elevated concentrations of particulate matter (PM_{2.5} and PM₁₀), nitrogen oxides, and black carbon compared to passenger vehicles. The spatial concentration of these emissions along freight corridors creates pollution hotspots that disproportionately affect communities, schools, and healthcare facilities in proximity to state logistics corridors a pattern documented in structural equation modeling of Brasília residents' quality of life, where noise and pollution from freight transport ranked among the most significant negative influences on urban livability (Gonzalez-Feliu et al., 2023). The study established that

19.5% of urban freight transport externalities were explained by public management, private management, and urban infrastructure factors.

In the PLI-SP 2050 context, these externality assessments connect directly to the State Climate Action Plan (PAC 2050), the State Climate Adaptation and Resilience Plan (PEARC), and the State Energy Plan 2050 (PEE 2050), which collectively establish the sustainability envelope within which freight network investments must operate (SEMIL, 2026). PAC 2050 sets sectoral emissions reduction targets requiring the freight sector to demonstrate measurable reductions in particulate matter and greenhouse gas emissions per tonne-kilometre transported. PEARC identifies infrastructure vulnerability to projected climate stressors including the urban heat island intensification documented along São Paulo's major freight corridors that will amplify the health consequences of freight-related air pollution for peripheral communities with limited adaptive capacity. PEE 2050's fleet electrification trajectory for commercial vehicles provides a long-horizon scenario framework for projecting how the emissions externality profile at urban nodes will evolve, informing the scenario analysis in SIMCU-State's policy formulation phase. The analytical framework proposed in this dissertation provides the urban-scale externality quantification necessary to evaluate whether PLI-SP 2050 corridor investment decisions generate net environmental improvements at urban nodes, accounting for induced demand effects, modal shift consequences, and the distributional pattern of emissions burdens documented by Cardoso (2025) across São Paulo's socioeconomic strata.

2.1.2 Economic Impacts: Congestion Costs and Delivery Efficiency

The economic impacts of freight-generated urban congestion operate across multiple scales with implications for PLI-SP 2050's cost-reduction objectives. At the macroeconomic level, logistics costs as a share of GDP in Brazil remain elevated relative to comparable economies due to

infrastructure deficiencies and operational inefficiencies that state-level investments aim to address. At the corridor scale, congestion increases delivery time variability, compelling carriers to maintain buffer time in scheduling that reduces vehicle utilization efficiency and elevates per-unit logistics costs that ultimately propagate through supply chains to affect consumer prices and industrial competitiveness (Soimun et al., 2025).

Research on a freight corridor in Indonesia demonstrated that volume-to-capacity ratios exceeding 1.02 during peak periods reduced freight vehicle speeds below 15 km/h, generating operating cost increases of 60% relative to free-flow conditions. Congestion costs reached the equivalent of IDR 30,879 per kilometer for large trucks in that study context, illustrating the order-of-magnitude economic penalties that inadequate corridor capacity imposes on freight operations (Soimun et al., 2025). Although direct cost transfer from Indonesian to Brazilian conditions requires local calibration, the mechanism volume-to-capacity ratios exceeding 1.0 compressing vehicle speeds and driving up operating costs applies universally to PLI-SP 2050 corridor analysis. A structurally analogous quantification using PLI-SP 2050 OD matrices weighted against Brazilian freight cost benchmarks from CNT (2022) employing the PLI-SP 2050 OD matrices to weight these unit costs by corridor freight volumes provides a quantitative foundation for infrastructure investment prioritization within the PLI-SP 2050 framework.

Taken together, the economic impact literature reveals a consistent pattern: congestion costs scale nonlinearly with volume-to-capacity ratios, and the absence of real-time freight exposure monitoring prevents practitioners from detecting the threshold conditions at which costs escalate most sharply precisely the monitoring gap SIMCU-State addresses. The Overall Equipment Effectiveness for Mobility (OEEM) indicator (Gonzalez-Feliu et al., 2023) encompasses delivery quality, vehicle utilisation, driver availability, traffic-related efficiency, and delivery stop

efficiency, whose compatibility with real-time dashboard visualisation makes it a candidate analytical module for SIMCU-State's operational reporting layer. The sustainability index for integrated urban transport and logistics developed by Marchetti and Wanke (2021) for São Paulo provides the state-level performance benchmark against which SIMCU-State's corridor-level monitoring outputs can be contextualised: their composite indicator incorporates modal integration quality, infrastructure adequacy, and last-mile delivery performance dimensions that directly correspond to the safety and congestion metrics SIMCU-State tracks in real time. Research on urban logistics and consumer behaviour in São Paulo by Dutra et al. (2021) characterises the demand-side dynamics e-commerce growth trajectories, just-in-time delivery frequency, and consumer proximity effects that determine urban node loading patterns at the distribution end of PLI-SP 2050 freight flows, providing the behavioural grounding for generation and attraction adjustments in SIMCU-State's DCEM disaggregation model.

2.1.3 Social Impacts: Quality of Life and Urban Livability

Urban freight's social impacts extend across dimensions of public space quality, community accessibility, and vulnerable road user safety that are increasingly central to state transport policy in São Paulo. The competition for curb space between freight loading operations, pedestrian movement, cycling infrastructure, and private vehicle parking generates conflicts that degrade public space functionality along commercial corridors conflicts that intensify as e-commerce delivery density increases along corridors connecting to PLI-SP 2050 logistics hubs (World Bank, 2023).

The barrier effect research conducted in São Carlos documented how high arterial traffic volumes create effective barriers to pedestrian and cyclist movement, with residents reporting dangerous conditions for active mode displacements on corridors handling significant freight volumes

(Guerin et al., 2021). This microlocal finding acquires systemic significance when situated within the structural inequality framework advanced by Cardoso (2025): the São Paulo Metropolitan Region's logistics geography concentrates the most intense freight activity precisely in those peripheral zones where public transport access is most limited, pedestrian infrastructure most deficient, and community political capacity to demand safety improvements lowest. The decentralisation of logistics warehousing documented by Guerin et al. (2021) driven by land cost differentials and highway access considerations has transferred the social costs of freight distribution from the central zone to peripheral municipalities along the SP-310, SP-330, and SP-348 corridors, creating geographically structured inequality in the freight externality burden that PLI-SP 2050's aggregate performance metrics do not capture. SIMCU-State's design explicitly addresses this gap by embedding distributional analysis across socioeconomic groups as a required output of its safety risk mapping, enabling PLI-SP 2050 to account for equity consequences alongside efficiency metrics in its investment prioritisation framework.

2.2 Road Safety Dimensions of Inter-Regional Freight at Urban Nodes

2.2.1 Freight Vehicle Involvement in Traffic Crashes at State Logistics Corridors

The road safety consequences of inter-regional freight vehicle movements at urban nodes represent one of the most policy-significant yet poorly quantified dimensions of state logistics planning. Research in North American freight corridors documented statistically significant increases in crashes involving commercial vehicles and vulnerable road users, with McDonald et al. (2019) reporting a 17% increase in fatalities for urban freight-related crashes compared to a 3% increase in overall traffic fatalities between 2009 and 2015. The spatial redistribution of freight crashes from intercity highways to local roads and arterials precisely the infrastructure at urban node entry

points where freight from state corridors disperses into urban distribution networks is documented as a significant and growing pattern.

A growing share of fatal truck accidents occurs in urban areas rather than on inter-urban highways, with the urban proportion increasing as e-commerce and last-mile freight activity intensify along state logistics corridors (McDonald et al., 2019). This spatial pattern is analytically critical for PLI-SP 2050 because it implies that state-level decisions about corridor routing, intermodal terminal placement, and urban bypass infrastructure directly determine the distribution of freight-related crash risk across urban nodes. The research framework proposed here enables this causal pathway to be modeled quantitatively, connecting PLI-SP 2050 planning decisions to urban safety outcome projections.

Research developing accident frequency prediction models for urban roads in São Paulo employed stepwise Poisson regression with variable selection, identifying junction density and commercial land use as statistically significant positive predictors of crash frequency (Isler et al., 2024). The predominance of directional and temporal clustering in crash data with 84.8% of accidents occurring in a single direction and 75.7% concentrated in a three-year period demonstrates heterogeneity patterns that state-urban integration models must accommodate through appropriate spatial and temporal stratification.

2.2.2 Vulnerable Road User Interactions with State Freight Flows

Interactions between inter-regional freight vehicles and vulnerable road users at urban nodes generate some of the most severe safety outcomes in the state's traffic system. Heavy vehicle blind spots extend substantial distances around large trucks and articulated combinations, limiting driver awareness of nearby pedestrians and cyclists at the complex intersections and loading areas that characterize urban node entry zones. Turning movements by heavy vehicles involve large swept

paths that frequently encroach on bicycle lanes and pedestrian crossings where urban road geometries were not designed to accommodate inter-regional freight vehicle dimensions (McDonald et al., 2019).

Posted speed limits and road cross-section design at urban node entry corridors significantly modulate severity outcomes. Posted speed limits and road geometry at urban node entry corridors significantly modulate injury severity outcomes, with higher operating speeds consistently associated with disproportionate increases in fatal injury probability for vulnerable road users (McDonald et al., 2019). Morning delivery hours (6 am to 12 pm) show significantly elevated severe injury odds, likely reflecting the convergence of inter-regional freight vehicle arrivals with urban peak commuting periods at node entry corridors. These temporal patterns, derivable from PLI-SP 2050 freight flow timing data, provide actionable intelligence for time-targeted safety interventions.

2.2.3 Driver Fatigue, Vehicle Maintenance, and Operational Safety Factors

The operational safety factors specific to inter-regional freight operations arriving at urban nodes differ meaningfully from those characterizing local urban delivery vehicles. Long-haul drivers arriving at urban nodes after extended highway travel may be experiencing fatigue accumulated over hundreds of kilometers of driving, creating elevated crash risk precisely at the complex navigational challenge of urban node entry. Tour-based freight operations, where drivers conduct multiple deliveries within urban areas after arriving from state corridors, generate distinctive fatigue exposure patterns with repeated stress-recovery cycles that differ analytically from continuous highway driving (Wang & Holguin-Veras, 2008).

The heterogeneity of the freight vehicle fleet serving São Paulo State's urban nodes is particularly relevant for safety analysis. Formal logistics operators employing modern vehicles with electronic

stability control, automatic emergency braking, and blind spot monitoring systems coexist with informal owner-operators maintaining aging vehicles without these safety technologies a heterogeneity that generates substantial variation in safety performance across otherwise identical traffic exposure conditions (CNT, 2022). The PLI-SP 2050 vehicle registration data available through DataGEO enables fleet composition characterization by corridor and node that supports differentiated safety modeling.

2.2.4 Spatial and Temporal Patterns of Freight-Related Incidents at Urban Nodes

Freight-related safety incidents at urban nodes exhibit spatial and temporal regularities that enable targeted analytical approaches when state-level freight flow data is integrated with urban crash records. Spatial concentration occurs at entry corridors from state highways, intermodal terminals, logistics warehouse districts, and the commercial cores that function as last-mile distribution destinations. Kernel density estimation and hotspot mapping techniques identify these concentrations from historical crash data, while PLI-SP 2050 OD matrices provide predictive freight exposure profiles for proactive risk mapping at nodes where historical crash data is sparse (Gomes et al., 2022).

The temporal dimension of freight safety patterns connects directly to PLI-SP 2050's planning for delivery time restriction policies. Off-hour delivery experiments in New York, Bogotá, and São Paulo demonstrated emissions reductions of 13%, 64%, and 48% respectively per kilometer traveled when goods flows were transferred to off-peak periods (Holguin-Veras et al., 2016). The extension of these findings to São Paulo State's urban node context where inter-regional freight arrivals and local delivery patterns overlap during morning peak hours provides an empirical foundation for PLI-SP 2050-time restriction recommendations that SIMCU-State can validate through before-after monitoring.

Research gap: The safety literature reviewed here characterises freight crash risk from locally collected traffic data. What is absent is a safety modelling framework that uses state-level freight demand estimates as the primary exposure metric, enabling prospective risk identification at urban nodes before crash accumulation is sufficient for historical modelling. This demand-driven, predictive safety modelling approach translating PLI-SP 2050 OD matrix flows into forward-looking crash risk assessments is the core contribution of Objective 3 and Research Question 3.

2.3 Traffic Performance Analysis at Freight Corridors

2.3.1 Level of Service Degradation from Inter-Regional Freight Movements

Inter-regional freight vehicles degrade traffic flow quality at urban nodes through mechanisms that reflect their physical, operational, and behavioral characteristics. Large vehicle presence reduces effective roadway capacity through size and slower acceleration rates, while operational behaviors including lane changes, turning maneuvers, and curb loading activities create turbulence in traffic streams that amplifies congestion propagation. Speed differentials between inter-regional freight vehicles and urban traffic generate shock waves whose spatial extent and duration depend on corridor geometry, traffic composition, and signal timing parameters that SIMCU-State's real-time monitoring architecture continuously tracks.

Research by Setti (2009) on speed-flow relationships for Brazilian expressways established that conventional Highway Capacity Manual classifications based on access control are inadequate for Brazilian conditions, with urban versus rural section distinction providing superior explanatory power. This finding implies that PLI-SP 2050's network performance modeling requires locally calibrated relationships rather than direct application of international standards a calibration requirement that SIMCU-State's real-time corridor monitoring data is designed to supply.

2.3.2 Network Reliability and Travel Time Variability

Travel time reliability at urban nodes on PLI-SP 2050 logistics corridors is a critical performance dimension for both inter-regional carrier operations and urban traffic management. Unreliable travel times at node entry corridors force carriers to maintain scheduling buffers that reduce vehicle productivity and constrain service offerings, with direct implications for PLI-SP 2050's logistics cost reduction objectives. Reliability metrics including the Planning Time Index, Buffer Time Index, and coefficient of variation quantify this uncertainty in forms directly applicable to PLI-SP 2050 performance monitoring frameworks (Soimun et al., 2025).

2.3.3 Intersection and Bottleneck Performance

Intersections at urban node entry points represent critical bottlenecks where inter-regional freight vehicles create disproportionate performance impacts. Signalized intersections handling heavy freight volumes require modified cycle lengths and phase structures to accommodate truck acceleration and clearance requirements while maintaining acceptable delay levels for local traffic. The spatial concentration of freight-related bottlenecks at specific intersection types where state highway traffic merges with urban arterials makes these locations priority targets for both SIMCU-State sensor deployment and PLI-SP 2050 infrastructure investment analysis (Setti, 2009).

Research gap: Traffic performance studies at Brazilian freight corridors rely on locally collected data and apply capacity standards calibrated to single-corridor contexts. No study has examined how state-level freight demand variations captured in periodic PLI-SP 2050 OD matrix updates propagate into performance degradation across multiple urban nodes simultaneously, or developed reliability metrics compatible with state logistics planning reporting. This multi-node, demand-driven performance monitoring is addressed by Objectives 4 and Research Questions 1 and 4.

2.4 Intelligent Transport Systems for State-Scale Freight Monitoring

2.5 Data-Driven Safety and Performance Modeling

2.5.1 Crash Frequency Prediction Models for Freight Exposure Analysis

Statistical modeling of crash frequency at urban freight nodes has matured around the Poisson and negative binomial regression framework, which accommodates the count nature of crash data and the overdispersion that routinely characterizes empirical distributions. The negative binomial formulation introduces a dispersion parameter that captures unobserved heterogeneity across sites heterogeneity that is particularly relevant when state OD matrix freight exposure estimates are used as predictors, since these estimates carry simulation-derived uncertainty not present in directly observed traffic counts (Lord & Mannering, 2010).

Research developing accident frequency prediction models for São Paulo urban roads employed stepwise Poisson regression with traffic volume (AADT), vehicle speed metrics, road geometry, land use characteristics, and intersection density as predictors (Isler et al., 2024). The integration of state freight exposure metrics derived from PLI-SP 2050 OD matrices as additional predictors extends this modeling framework to capture the contribution of inter-regional freight flows to urban node crash risk a contribution that conventional models based solely on local traffic counts cannot isolate.

2.5.2 Before-After Study Designs for PLI-SP 2050 Evaluation

Before-after study designs provide the most rigorous quasi-experimental framework for evaluating PLI-SP 2050 infrastructure and regulatory interventions at urban nodes. Simple before-after comparisons risk confounding from concurrent trends economic cycles, secular safety improvements, weather variation that affect outcomes independently of the intervention. Comparison group methods control for these confounders by contrasting treated urban nodes with

matched untreated nodes experiencing similar external conditions drawn from the PLI-SP 2050 corridor typology (Lord & Mannering, 2010).

The Empirical Bayes method combining observed crash frequencies with predictions from reference safety performance functions provides refined before-after estimates by reducing regression-to-mean bias that distorts simple comparisons. Research on signal countdown devices across Brazilian cities demonstrated the Empirical Bayes methodology, finding crash reductions of 30 to 45% associated with specific intersection safety treatments (Hauer et al., 2002; Lord & Mannering, 2010). Application of comparable methodology to PLI-SP 2050 corridor interventions provides an evaluative framework whose statistical properties are well-suited to the spatial scale and data availability of state logistics network analysis.

2.5.3 State-Level OD Matrix Disaggregation for Urban Node Analysis

The disaggregation of state-level origin-destination matrices into urban node traffic loading profiles is a methodologically novel challenge that combines established freight demand modeling techniques with the specific data properties of SEMIL's 62 OD matrices. Land use-based freight trip generation models provide the theoretical foundation, establishing statistical relationships between freight activity and observable establishment characteristics that enable demand inference from land use data available through DataGEO (Gomes et al., 2022).

Entropy maximization methods offer a theoretically grounded approach to disaggregating zonal OD flows into urban network loadings that are consistent with both the state-level demand estimates and the capacity constraints of urban node infrastructure. The tour-based entropy maximization formulations of Wang and Holguin-Veras (2008) address the chained nature of freight vehicle movements within urban distribution zones movements that violate the independence assumptions underlying conventional OD-based approaches providing a

methodologically rigorous disaggregation framework compatible with SEMIL's multimodal matrix structure.

The academic literature engaged by SIMCU-State reflects Prof. André's macro-to-micro analytical logic, and is sequenced accordingly. At the macro socioeconomic layer, Cardoso (2025) on logistics and inequality in the São Paulo Metropolitan Region constitutes the lead reference for understanding how PLI-SP 2050's freight network operates within and reproduces São Paulo's centro-periphery spatial inequality. Cardoso's documentation of asymmetric externality distribution across ring and radial corridors establishes the equity baseline against which SIMCU-State's distributional safety and congestion analysis is calibrated, and provides the socioeconomic geography framework that grounds the Socioeconomic Characterisation layers within DataGEO.

3. RESEARCH PROBLEM STATEMENT

3.1 Core Problem Definition

3.1.1 The State-Urban Translation Gap in Logistics Planning

The fundamental problem addressed by this research is the absence of a validated analytical framework capable of translating São Paulo State's macro-level freight planning intelligence as embodied in PLI-SP 2050's OD matrices, socioeconomic characterization, and multimodal network models into actionable urban node impact assessments for road safety and traffic performance. SEMIL's DataGEO platform and the 62 OD matrices represent an unprecedented state-level data asset, yet this asset's utility for urban traffic management remains unrealized because no systematic methodology exists for disaggregating state freight demand estimates into the local-scale traffic loading profiles necessary for safety and performance modeling.

This translation gap has concrete operational consequences for PLI-SP 2050 implementation. Infrastructure investment decisions affecting freight corridor capacity, intermodal terminal placement, and urban bypass routing are made with limited knowledge of their urban node safety and congestion implications. Regulatory decisions affecting time-of-day delivery restrictions, vehicle type limitations, and access control at urban nodes cannot be evaluated for their network-wide consequences without models that connect state-level demand to urban traffic outcomes. Policy assessments of PLI-SP 2050 interventions consequently remain incomplete analytically unable to account for the externality costs imposed on urban populations by state logistics decisions.

3.1.2 Reactive Rather Than Proactive Urban Node Management

The absence of a real-time state-urban freight monitoring capability constrains urban node management to reactive approaches that respond to incidents and congestion after they have

developed rather than anticipating their emergence from upstream freight flow data. PLI-SP 2050's OD matrices provide the latent capability to forecast urban node loading profiles based on state corridor demand yet this forecasting potential is not operationalized in current urban traffic management practice because the technical infrastructure for connecting state freight demand data to urban monitoring systems does not exist.

The temporal concentration of inter-regional freight arrivals at urban nodes shaped by corridor distance, driver scheduling practices, and state highway travel time distributions is knowable in principle from PLI-SP 2050 corridor data, yet it is not currently used to pre-position traffic management resources or adjust signal timing in anticipation of elevated freight loads. The morning period (6 am to 12 pm) that consistently generates elevated severe crash risk in urban freight research (McDonald et al., 2019) is precisely the period when inter-regional freight vehicles complete overnight highway journeys and enter urban distribution zones, creating a predictable convergence of elevated exposure and elevated risk that reactive management approaches address only after incidents have occurred.

3.1.3 Data Fragmentation Across State and Municipal Institutional Silos

Freight-relevant data for urban node analysis resides in institutionally disconnected systems across multiple governmental levels. At the state level, SEMIL's DataGEO platform consolidates logistics, infrastructure, and environmental data but integration with urban-level accident records, traffic flow measurements, and land use data held by municipal agencies requires deliberate architectural effort. At the municipal level, public security agencies hold accident records, transportation departments maintain traffic counts, motor vehicle departments administer vehicle registration, and planning departments manage land use data with different spatial referencing

systems, temporal aggregation conventions, and access restriction frameworks that impede integrated analysis.

The technical and institutional barriers to cross-level data integration are compounded by São Paulo's well-documented fragmented governance architecture. Transport data in the São Paulo State context is distributed not only across state and municipal agencies but also across formal and informal economic sectors whose freight activities are captured at different rates in official statistics. PLI-SP 2050's OD matrices, constructed from formal logistics operators, registered establishments, and DER-SP classified traffic counts, systematically underrepresent the informal freight economy including the motoboy delivery networks, informal vehicle operators, and unregistered commercial distribution patterns that constitute a significant share of last-mile freight activity in peripheral municipalities (Dutra et al., 2021). This data fragmentation is not merely a technical measurement problem but a structural governance challenge: informal freight operations occupy the same road infrastructure as formal inter-regional freight at urban nodes, generating safety risks and congestion that the formal data architecture cannot see. The research team that developed São Paulo crash prediction models explicitly documented this challenge, requiring substantial preprocessing to achieve spatial alignment among CET crash data, roadside camera speed data, and Geosampa land use data (Isler et al., 2024). SIMCU-State's architecture is specifically designed to operationalise the institutional partnership between SEMIL and municipal transport agencies established through PLI-SP 2050's development process, while explicitly documenting the informal economy data gap as a known uncertainty boundary in its safety and performance model specifications.

3.2 Specific Challenges in State-Level Freight-Urban Integration

3.2.1 OD Matrix Spatial Resolution and Urban Node Disaggregation

SEMIL's 62 OD matrices characterize freight flows at a spatial resolution appropriate for state-scale network modeling but requiring disaggregation for urban node impact analysis. The translation from inter-zonal flow estimates to specific urban road link loading profiles involves spatial allocation procedures that propagate uncertainty from the state-level demand estimation through to urban safety and performance predictions. Calibrating these disaggregation procedures against observed urban traffic data available through municipal sensor networks and the camera infrastructure at urban nodes is the central methodological challenge for state-urban freight impact modeling.

3.2.2 Multimodal Data Integration for São Paulo State Freight Characterization

PLI-SP 2050's multimodal framework encompasses road, rail, waterway, and pipeline freight movements whose interactions at multimodal terminals and urban nodes generate traffic impacts that unimodal analysis cannot adequately characterize. The disaggregation of multimodal OD matrices into urban node traffic loading profiles requires assumptions about mode-specific temporal distributions, urban access routing, and terminal dwell time patterns that are not directly observable from state-level data. The incorporation of SEMIL's multimodal matrix structure into SIMCU-State's analytical framework represents a methodological advance over existing urban freight monitoring systems that address only road freight without accounting for modal interchange dynamics.

3.2.3 Mixed Traffic Conditions with High Vulnerable Road User Exposure

São Paulo State's urban environments are characterized by the extreme traffic heterogeneity that typifies Brazilian cities heavy freight vehicles sharing road infrastructure with substantial

motorcycle, pedestrian, and bicycle volumes in circumstances where inter-regional freight vehicles are navigating unfamiliar urban environments after extended highway journeys. The mortality data for São Paulo State, where pedestrians constituted 31.82% and motorcycle occupants 29.30% of traffic fatalities between 2000 and 2016 (Aquino et al., 2020), illustrates the disproportionate risk exposure of vulnerable road users at urban nodes where state freight corridor traffic disperses into local distribution networks.

3.3 Research Questions

3.3.1 Primary Research Question

How can an integrated real-time Intelligent Transport System architecture be designed, implemented, and validated to operationalise São Paulo State's PLI-SP 2050 inter-regional freight flow data as actionable urban node impact assessments quantifying road safety risks and traffic performance degradation that support proactive urban traffic management and evidence-based state logistics policy development?

This overarching question encompasses: the technical and institutional feasibility of state-urban data integration through SEMIL's DataGEO platform; the analytical validity of OD matrix disaggregation methodologies for urban node traffic loading estimation; the predictive accuracy of safety and performance models incorporating state freight exposure metrics; and the operational utility of SIMCU-State's output products for both municipal traffic management and PLI-SP 2050 policy workflows.

3.3.2 Secondary Research Questions

SRQ1: What disaggregation methodologies provide optimal accuracy in translating SEMIL's 62 OD matrices into urban node traffic loading profiles, and how do prediction errors propagate from state-level demand estimation through to urban safety and performance model outputs?

SRQ2: How accurately can inter-regional freight trip patterns at urban nodes be estimated using PLI-SP 2050 socioeconomic characterization and land use data combined with entropy-based distribution methods calibrated against municipal traffic counts?

SRQ3: What statistical model specifications best predict crash frequencies and severities at urban nodes incorporating state-level freight exposure metrics from PLI-SP 2050 OD matrices, and how do their predictive performances compare to models using only locally observed traffic data?

SRQ4: What traffic performance indicators level of service, travel time reliability, intersection delay are most sensitive to inter-regional freight flow variations at urban nodes, and how can they be monitored in real-time through SEMIL DataGEO integration?

SRQ5: What technical standards and institutional arrangements enable effective data sharing between SEMIL's state-level DataGEO platform and municipal traffic management systems, while maintaining the data quality and access restrictions appropriate to each institutional context?

SRQ6: How do safety and performance outcomes at urban nodes with SIMCU-State-enabled management compare to matched control nodes across PLI-SP 2050 corridor typologies, and what effect sizes are statistically detectable with the available monitoring data?

SRQ7: What PLI-SP 2050 policy interventions modal shift incentives, bypass infrastructure investments, urban access regulations, time-of-day delivery restrictions are most cost-effective for improving safety and performance outcomes at state logistics urban nodes, and how do optimal designs vary across node typologies?

4. RESEARCH OBJECTIVES

4.1 Primary Objective

Develop, implement, and validate SIMCU-State an integrated Intelligent Transport System architecture for real-time monitoring of inter-regional freight flow impacts on road safety and traffic performance at São Paulo State urban nodes demonstrating technical feasibility, analytical validity, and operational utility through implementation aligned with PLI-SP 2050's state logistics planning framework and validated against SEMIL's multimodal origin-destination matrices.

4.2 Specific Objectives

4.2.1 Objective 1: Characterize State-Urban Freight Patterns at São Paulo Logistics Nodes

This objective addresses the foundational analytical challenge of disaggregating state-level freight demand data into urban node traffic loading profiles. The 62 OD matrices provided by SEMIL for PLI-SP 2050 characterize inter-zonal flows at a spatial resolution suitable for state network modeling, requiring systematic disaggregation procedures validated against observed urban traffic data. The PLI-SP 2050 socioeconomic characterization provides establishment size, employment, and economic activity data by zone that supplements the OD matrix information with land use-based freight generation estimates.

Components include: (1) State OD Matrix Analysis extract freight flow volumes, commodity compositions, and temporal profiles from SEMIL's 62 matrices relevant to target urban node corridors; (2) Urban Network Loading Estimation develop and validate spatial disaggregation procedures for translating inter-zonal flows into urban road link volumes; (3) Land Use Integration incorporate PLI-SP 2050 socioeconomic characterization data to refine freight generation and attraction estimates at node destinations; and (4) Validation compare estimated loading profiles

against observed traffic counts at urban node entry corridors and INFOSIGA-SP vehicle classification records.

4.2.2 Objective 2: Design State-Urban Integrated Data Architecture

The architecture design connects the three-layer ITS model IoT device layer, platform and integration layer, and BI visualization layer with a fourth state integration layer that interfaces SIMCU-State with SEMIL's DataGEO platform. This state integration layer constitutes the architecturally novel contribution of SIMCU-State relative to conventional city-scale ITS deployments, providing the technical foundation for real-time calibration of state OD matrix estimates against urban monitoring observations.

Components include: (1) DataGEO API Integration establish automated data exchange protocols connecting SIMCU-State's analytical pipeline to SEMIL's DataGEO platform layers; (2) IoT Sensor Network deploy vehicle detection and classification sensors at strategic urban node entry corridors identified through PLI-SP 2050 freight flow analysis; (3) Video Analytics Integration configure existing camera networks at urban nodes with automated vehicle detection and classification; (4) State Database Linkage establish secure data pipelines from INFOSIGA-SP crash records, DETRAN-SP vehicle registration, and PLI-SP 2050 socioeconomic characterization; and (5) Data Fusion Engine implement quality control, temporal synchronization, and cross-platform integration algorithms.

4.2.3 Objective 3: Analyze Road Safety Impacts at Urban Nodes

The integration of PLI-SP 2050 freight exposure metrics into predictive safety models enables identification of emerging risk conditions before crash accumulation, representing a substantial methodological advance over models calibrated exclusively on historical local traffic data. State OD matrix data provides freight exposure estimates for urban node corridors that extend the

temporal horizon of safety modeling beyond the retrospective window available from municipal crash records.

Components include: (1) Crash Frequency Modelling develop Poisson and negative binomial regression models predicting crash counts as functions of state freight exposure metrics, local traffic conditions, infrastructure characteristics, and PLI-SP 2050 Socioeconomic Characterisation zone indicators (employment formality rate and income distribution by planning zone) as structural predictors capturing differential vulnerability across São Paulo's centro-periphery axis; (2) Spatial Risk Mapping apply kernel density estimation to identify and visualise safety hotspot patterns at urban nodes, cross-referenced with PLI-SP 2050 corridor typologies and ZEE-SP land use zoning, enabling distributional analysis of risk concentration across socioeconomic geographies as documented by Cardoso (2025); (3) Freight Vehicle Involvement Analysis quantify inter-regional freight vehicle involvement in severe and fatal crashes relative to state OD matrix exposure estimates, stratified by node socioeconomic typology; and (4) Vulnerable Road User Risk model pedestrian and cyclist crash risks in state freight-intensive urban corridors using PLI-SP 2050 socioeconomic characterisation data as structural predictors, producing distributional risk profiles that enable PLI-SP 2050 to target safety interventions toward nodes where freight exposure and socioeconomic vulnerability coincide most acutely, advancing São Paulo State's SDG 11 commitments to inclusive and safe urban mobility.

4.2.4 Objective 4: Assess Traffic Performance Impacts at State Logistics Nodes

Traffic performance assessment at PLI-SP 2050 urban nodes requires adaptation of standard analytical frameworks to the specific conditions of Brazilian inter-regional freight corridors, where international capacity standards require local calibration to reflect the speed-flow relationships documented for Brazilian expressways (Setti, 2009).

Components include: (1) Corridor-Level Assessment calculate volume-to-capacity ratios and level of service for urban node entry corridors identified through PLI-SP 2050 freight flow analysis; (2) Reliability Measurement measure travel time variability using Planning Time Index and buffer time metrics calibrated against PLI-SP 2050 logistics cost reduction targets; (3) Intersection Analysis evaluate intersection delay and queue formation at node entry points with freight-specific phase timing analysis; and (4) Network Resilience assess urban node network responses to corridor disruptions and inter-regional freight demand variations using PLI-SP 2050 scenario data.

4.2.5 Objective 5: Implement Real-Time Monitoring Dashboard for PLI-SP 2050 Integration

The translation of SIMCU-State's analytical outputs into operational intelligence for both municipal traffic management and PLI-SP 2050 policy workflows requires explicit attention to the dual user context urban traffic operators requiring real-time incident response support and state freight planners requiring corridor performance metrics compatible with PLI-SP 2050 reporting frameworks.

Components include: (1) DataGEO-Compatible Visualization develop dashboard outputs formatted for integration with SEMIL's existing DataGEO visualization infrastructure; (2) Municipal Traffic Management Interface develop real-time display for traffic management centers showing urban node freight conditions and emerging safety risks; (3) Alert Generation configure threshold-based and anomaly detection systems for both urban incident response and PLI-SP 2050 performance threshold monitoring; and (4) Decision Support enable drill-down investigation and scenario analysis capabilities for both municipal operators and state planners.

4.2.6 Objective 6: Validate System Effectiveness Across PLI-SP 2050 Node Typologies

Validation of SIMCU-State across multiple urban node typologies drawn from the PLI-SP 2050 node classification framework tests the generalizability of the state-urban translation methodology beyond a single case study site, substantially strengthening the research's contribution to PLI-SP 2050's analytical toolkit.

Components include: (1) Multi-Node Before-After Design define pre-implementation and post-implementation observation periods across urban nodes of distinct PLI-SP 2050 typologies; (2) Comparison Node Selection identify matched control urban nodes from PLI-SP 2050 corridor typologies for causal inference; (3) Predictive Accuracy Testing assess OD matrix disaggregation model performance against holdout traffic count data at validation nodes; and (4) Sensitivity Analysis evaluate model robustness to variations in state OD matrix specification and municipal data availability across node contexts.

4.2.7 Objective 7: Formulate Evidence-Based Policy Recommendations for PLI-SP 2050

The development of policy recommendations aligned with PLI-SP 2050's strategic objectives logistics cost reduction, multimodal integration, environmental sustainability, and road safety improvement ensures research utility extends from academic contribution to direct support for state logistics governance.

Components include: (1) PLI-SP 2050 Scenario Analysis simulate effects of alternative state infrastructure investments and regulatory interventions using SIMCU-State validated models; (2) State-Municipal Stakeholder Engagement conduct structured workshops with SEMIL planners, municipal traffic management agencies, logistics operators, and community representatives; (3) Policy Formulation translate analytical findings into recommendations for PLI-SP 2050 urban node management guidelines; and (4) Replication Guidelines document methodological

adaptations for extending SIMCU-State to additional São Paulo State urban nodes with varying data availability and infrastructure contexts.

5. RESEARCH METHODOLOGY

5.1 Phase 1: Foundation and Data Preparation

5.1.1 State OD Matrix Integration and Urban Network Coding

The initial phase establishes the analytical foundation by integrating SEMIL's PLI-SP 2050 datasets with urban transport network models. The zone delineation for urban node analysis must balance: alignment with PLI-SP 2050 spatial units for OD matrix disaggregation; internal homogeneity in freight characteristics for model estimation reliability; compatibility with INFOSIGA-SP geocoding for crash record integration; and nested structure enabling aggregation for state-level reporting and disaggregation for urban-level analysis.

Steps include: (1) Extract and process relevant OD matrices from SEMIL's 62-matrix suite, identifying those covering target urban nodes and primary logistics corridors; (2) Delineate traffic analysis zones (60-120 zones per urban node) aligned with PLI-SP 2050 spatial units; (3) Code road networks with functional classification, topology, and attribute databases incorporating PLI-SP 2050 infrastructure characterization data; (4) Identify freight corridor entry points at urban nodes as priority locations for sensor deployment; and (5) Validate spatial alignment across DataGEO geospatial layers, INFOSIGA-SP crash geocoding, and urban network models.

5.1.2 State-to-Urban Freight Demand Disaggregation

The methodological core of Phase 1 is the development and calibration of procedures for disaggregating SEMIL's inter-zonal OD matrices into urban node traffic loading profiles. The entropy maximization formulation maximizes distribution entropy subject to zonal totals from state OD matrices, observed link flow constraints from urban sensor networks, and impedance functions calibrated to local network conditions following the tour-based entropy framework of Wang and Holguin-Veras (2008) adapted for PLI-SP 2050's multimodal data structure.

Steps include: (1) State OD matrix processing extract inter-zonal flows, commodity compositions, and temporal distributions for target urban nodes; (2) PLI-SP 2050 socioeconomic data integration apply land use-based generation and attraction adjustments using characterization data from DataGEO; (3) Temporal distribution estimation derive hourly freight arrival profiles at urban node entry corridors from state corridor travel time distributions and carrier scheduling patterns; (4) Entropy-based urban network loading synthesis with state OD totals, network impedances, and available count calibration; and (5) Validation against observed urban traffic counts, DETRAN-SP vehicle registration classifications, and available carrier interview data.

5.2 Phase 2: SIMCU-State System Development

5.2.1 Architecture Design with State Integration Layer

The SIMCU-State architecture extends the conventional three-layer ITS model with a state integration layer: Edge Layer with sensor interfaces, local processing, and quality control targeting sub-second latency; Platform Layer with data ingestion, storage, indexing, and API services; Analytics Layer with OD matrix disaggregation engine, safety and performance model execution, and prediction generation; State Integration Layer with DataGEO API connectors, PLI-SP 2050 matrix update protocols, and SEMIL reporting interfaces; and Presentation Layer with dual-output dashboards for municipal traffic management and state logistics planning audiences.

5.2.2 Data Fusion Engine with State-Municipal Integration

The data fusion engine integrates real-time urban monitoring data with PLI-SP 2050 state data streams: quality control with range checks, temporal consistency, and cross-sensor validation targeting less than 5% false positive anomaly rate; timestamp synchronization using network time protocol and GPS reference achieving less than 100 milliseconds synchronization error; vehicle identity resolution using probabilistic matching and trajectory continuity targeting greater than

90% correct re-identification; state OD matrix update protocols managing the integration of periodic matrix revisions with real-time monitoring data; and missing data handling with imputation procedures and uncertainty propagation for valid inference despite typical 10-20% sensor missingness rates.

5.3 Phase 3: Safety Impact Analysis at Urban Nodes

5.3.1 Crash Frequency Modeling with State Freight Exposure

The model specification incorporates PLI-SP 2050 state freight exposure as the primary novel predictor class alongside conventional traffic, infrastructure, and socioeconomic vulnerability variables. Dependent variable specification uses crash counts by urban node corridor segment and time period, employing Poisson or negative binomial distributions. The predictor set includes: state OD matrix freight exposure (vehicles per hour by vehicle category, derived from DCEM disaggregation); total AADT; intersection density; land use mix from PLI-SP 2050 characterisation; PLI-SP 2050 Socioeconomic Characterisation zone indicators including employment formality rate and income quintile distribution by planning zone included as structural predictors to capture the differential vulnerability of communities across São Paulo's centro-periphery axis, following the distributional logic advanced by Cardoso (2025); ZEE-SP ecological-economic zone classification for land use heterogeneity and environmental sensitivity; and temporal indicators for seasonal freight demand cycles captured in PLI-SP 2050's matrix temporal dimension, including agricultural commodity peaks on the SP-310 and SP-330 corridors documented by Caixeta-Filho et al. (2022). Model selection employs AIC, BIC, cross-validation, and overdispersion testing with the negative binomial preferred when overdispersion is present (Lord & Mannering, 2010). Where significant spatial autocorrelation is detected in model residuals (Moran's I test, $p < 0.05$), a Spatial Durbin Model (SDM) or geographically weighted regression

(GWR) specification will be substituted to account for spatial dependence structures that standard GLMs cannot capture. All statistical modelling will be implemented in R (version 4.3+) using the MASS, spdep, and GWmodel packages; spatial analysis in ArcGIS Pro with Network Analyst; traffic demand modelling in TransCAD; and the real-time data pipeline in Python 3.11+ with FastAPI, PostgreSQL/PostGIS, and Apache Kafka.

5.3.2 Spatial Risk Mapping Across PLI-SP 2050 Node Typologies

Techniques include: kernel density estimation for continuous risk surface generation at urban nodes, with bandwidth optimization cross-validated against INFOSIGA-SP crash distribution patterns; hotspot identification using threshold-based priority location selection validated across PLI-SP 2050 node typologies; and spatial regression connecting safety risk surfaces to PLI-SP 2050 freight flow intensity metrics using spatial autocorrelation assessment.

5.4 Phase 4: Traffic Performance Analysis

5.4.1 Corridor-Level Assessment at PLI-SP 2050 Urban Nodes

Metrics include: volume-to-capacity ratio calculated from observed flows and capacity adjusted for inter-regional freight passenger car equivalents, calibrated against Setti's (2009) Brazilian speed-flow relationships; level of service using adapted HCM classifications for Brazilian urban conditions; travel time reliability indices including Planning Time Index and Buffer Time Index benchmarked against PLI-SP 2050 logistics cost reduction targets; and freight-specific delay quantifying excess travel time attributable to inter-regional freight vehicle presence at urban node entry corridors.

5.5 Phase 5: System Validation and Evaluation

5.5.1 Before-After Study Design Across PLI-SP 2050 Node Typologies

The study design employs: pre-implementation periods of minimum 12 months at each urban node for adequate crash accumulation and seasonal coverage; post-implementation periods of minimum 12 months, targeting 24 or more months for sustained effect detection; comparison node selection from matched PLI-SP 2050 corridor typologies to control for state-level freight demand trends; and statistical power targeting 80% to detect 20% reductions in crash frequency the minimum policy-relevant effect size at node sample sizes derivable from PLI-SP 2050's urban node inventory.

5.6 Phase 6: Policy Formulation and Dissemination

5.6.1 PLI-SP 2050 Scenario Analysis

Intervention categories for scenario analysis, aligned with PLI-SP 2050 strategic options and evaluated against SDG 9 and SDG 11 performance criteria, include: multimodal shift incentives evaluated through network simulation of freight redistribution from road to rail or waterway on state corridors, with urban node traffic, safety, and emissions consequence modelling aligned with PAC 2050 decarbonisation targets (Caixeta-Filho et al., 2022; Vieira et al., 2016); urban bypass infrastructure assessed through network loading projections for proposed freight corridor realignments in PLI-SP 2050's infrastructure investment portfolio, with distributional analysis of safety and congestion impact transfers across socioeconomic zones informed by Cardoso's (2025) centro-periphery inequality geography; time-of-day delivery regulations evaluated through discrete choice modelling of carrier scheduling responses, with explicit modelling of informal freight operator compliance capacity a critical gap in conventional regulation impact assessment identified by Dutra et al. (2021); fleet electrification transition scenarios modelled against PEE 2050's commercial vehicle electrification trajectory, assessing how changing fleet composition

alters noise, emissions, and safety profiles at urban nodes over the 2030–2050 horizon; and climate resilience interventions aligned with PEARC scenario parameters, evaluating how projected climate stressors affect freight corridor reliability and urban node safety outcomes under high and low adaptation investment scenarios.

5.6.2 Stakeholder Engagement

Stakeholder groups and engagement mechanisms include: SEMIL planners and PLI-SP 2050 development team through technical workshops on state-urban translation methodology and DataGEO integration architecture; municipal traffic management center operators through system demonstration, workflow integration protocols, and operational feedback collection; logistics carriers and shippers operating on PLI-SP 2050 state corridors through industry forums and cost-benefit assessment of urban node management improvements; community representatives at major urban nodes through public meetings and quality of life impact assessment; and state and municipal policymakers through briefings, PLI-SP 2050 policy option analysis, and implementation planning sessions.

5.7 Research Timeline and Milestone Plan

The research is designed for completion within the standard EESC-USP PhD timeline of 48 months, with a maximum extension of 56 months if post-implementation monitoring data requires additional collection time. The six research phases are distributed across three consecutive years with deliberate phase overlap.

Year 1, Semesters 1–2 (Months 1–12): Foundation and Architecture. Semester 1 encompasses PhD qualifying examination, formalisation of the SEMIL data research agreement, extraction and processing of PLI-SP 2050 OD matrices, traffic analysis zone delineation for São Carlos, and initial network coding. Key milestone: signed SEMIL data access protocol and validated zoning

framework. Semester 2 encompasses DCEM disaggregation model development and calibration, SIMCU-State architecture design with DataGEO API prototype, IoT sensor deployment at São Carlos, and commencement of pre-implementation monitoring. Key milestone: calibrated disaggregation model validated against municipal traffic counts; first peer-reviewed manuscript submitted.

Year 2, Semesters 3–4 (Months 13–24): Analysis and Modelling. Semester 3: INFOSIGA-SP crash data extraction and geocoding; negative binomial safety model estimation with state freight exposure metrics; spatial hotspot mapping; preliminary traffic performance assessment. Key milestone: submitted journal paper on disaggregation methodology and safety modelling. Semester 4: full traffic performance analysis; extension to a second PLI-SP 2050 validation node; commencement of post-implementation monitoring at São Carlos. Key milestone: Exame de Qualificação passed; second journal paper submitted on traffic performance framework.

Year 3, Semesters 5–6 (Months 25–48): Validation, Policy, and Thesis. Semester 5: DiD validation analysis, sensitivity analysis, PLI-SP 2050 scenario analysis, and stakeholder workshops. Semester 6: thesis writing, a third peer-reviewed publication on validation results and policy recommendations, replication guideline documentation, and thesis defence preparation. Minimum viable dissertation scope: validated DCEM methodology, safety model with state exposure metrics, SIMCU-State architecture prototype, and scenario analysis achievable within 48 months regardless of post-implementation monitoring timeline.

5.8 Ethical Considerations and Data Governance

The research involves collection and processing of personal data through traffic camera networks and vehicle identification systems. In Brazil, the Lei Geral de Proteção de Dados Pessoais (LGPD, Law 13.709/2018) governs the processing of personal data, requiring a legal basis, purpose limitation, and data minimisation. SIMCU-State's sensor data collection will be designed to capture vehicle type, speed, and classification without individual licence plate retention beyond the minimum period required for trajectory analysis. All data sharing between SEMIL, municipal agencies, and EESC-USP will be governed by formal data use agreements specifying retention periods, access controls, and anonymisation protocols. Approval from the EESC-USP Research Ethics Committee (CEP) will be sought prior to human participant engagement in stakeholder workshops and carrier interviews. Triangulation of findings will be achieved through convergence across three data streams: quantitative safety and performance model outputs from INFOSIGA-SP and SEMIL data; real-time sensor monitoring observations from SIMCU-State deployments; and qualitative stakeholder perspectives from structured workshops. Researcher positionality is acknowledged: as a doctoral candidate at EESC-USP with institutional proximity to the SEMIL/PLI-SP 2050 research partnership, the researcher holds a position of relative insider access that facilitates data collection but requires reflexive attention to confirmation bias in interpreting policy-relevant findings. This is mitigated through pre-registered analysis protocols, external peer review of all three target publications, and transparent reporting of model limitations. Community consultation at urban node sites will follow the participatory principles established in Brazil's National Urban Mobility Policy (Law 12.587/2012).

6. DATA SOURCES: PLI-SP 2050 AND SEMIL DATASETS

6.1 Rationale for State-Level Data Integration

6.1.1 Advantages of SEMIL's Centralized Data Infrastructure

The rationale for centering SIMCU-State's data architecture on SEMIL's state-level infrastructure rather than relying primarily on municipal sensor networks and periodic local surveys reflects both the analytical advantages of comprehensive state data and the institutional realities of freight research in Brazilian medium-sized cities. Municipal-level freight data collection faces persistent barriers: limited technical capacity, funding constraints, jurisdictional fragmentation, and the institutional resistance to data sharing that characterizes many Brazilian municipal agencies. SEMIL's DataGEO platform circumvents these barriers by providing centralized access to state geospatial data across institutional boundaries, creating a stable analytical foundation that does not depend on case-by-case municipal data sharing agreements.

The 62 OD matrices assembled for PLI-SP 2050 represent a data asset of unmatched scope for Brazilian state freight research. Covering the full São Paulo State multimodal network for the base planning year, these matrices provide freight flow characterization at spatial and modal resolution that would require decades of incremental municipal surveys to replicate from the bottom up. Their integration into SIMCU-State's analytical framework transforms a potentially data-scarce research environment into one of the best-documented state freight systems in Latin America a transformation that substantially amplifies the research's analytical scope and policy relevance.

6.1.2 Representative Urban Node Selection Within PLI-SP 2050 Framework

Urban node selection for SIMCU-State implementation and validation is guided by PLI-SP 2050's node typology framework rather than single-city representativeness considerations. This typology-based selection strategy produces a validation dataset spanning the diversity of freight-urban

interaction patterns in São Paulo State, enabling generalization of SIMCU-State's methodology across the full range of PLI-SP 2050's urban node inventory rather than to a single comparable city.

Key characteristics of candidate urban nodes include: position in the PLI-SP 2050 logistics network hierarchy (primary hub, secondary distribution center, corridor waypoint); freight commodity composition and modal characteristics derived from OD matrix analysis; existing monitoring infrastructure availability at potential sensor deployment sites; institutional relationships with SEMIL and receptiveness to PLI-SP 2050 research collaboration; and INFOSIGA-SP crash data availability and geocoding quality for safety model calibration. The São Carlos case is retained as an initial implementation site given its existing institutional relationships with the University of São Paulo and its established research infrastructure, while additional validation nodes are selected from PLI-SP 2050's interior corridor inventory.

6.1.3 SEMIL DataGEO Platform and PLI-SP 2050 Information Bank

The DataGEO platform centralizes São Paulo State's geospatial information across environmental, infrastructure, and logistics domains, providing the foundational spatial data infrastructure for SIMCU-State's analytical pipeline. Available layers relevant to SIMCU-State include: PLI-SP 2050 logistics network characterization with corridor classifications, capacity data, and freight flow estimates; state highway infrastructure condition and geometric data for corridor performance modeling; Ecological-Economic Zoning (ZEE-SP) land use classifications for freight generation and attraction estimation; environmental monitoring data for emissions and noise impact assessment aligned with PAC 2050 targets; and climate scenario layers from PEARC for resilience analysis of freight corridors under projected climate conditions (SEMIL, 2026).

The PLI-SP 2050 information bank provides supplementary analytical resources including socioeconomic characterization of São Paulo State municipalities, multi-year series data for freight demand trend analysis, and interactive dashboard tools for state-level network performance visualization. These resources support both the calibration of SIMCU-State's analytical models and the integration of research outputs into PLI-SP 2050's ongoing planning workflows.

6.1.3 SEMIL DataGEO Platform and PLI-SP 2050 Information Bank

6.1.4 Data Access Strategy and Risk Mitigation

Access to SEMIL's PLI-SP 2050 OD matrices, DataGEO platform APIs, and INFOSIGA-SP crash records requires formal research agreements between EESC-USP and the relevant state secretariats. The research treats formalisation of these agreements as a Phase 0 prerequisite, pursued during the first semester in parallel with literature review activities. Preliminary engagement with SEMIL has been initiated through the EESC transportation engineering group's existing institutional relationship with the PLI-SP 2050 planning team. A formal letter of intent from SEMIL confirming data availability and access terms will be sought prior to the PhD admission examination.

Three data access risk scenarios are anticipated with contingency responses. First, if full OD matrix access is delayed beyond Month 6, the disaggregation methodology will be developed using publicly available DER-SP corridor traffic counts and DETRAN-SP vehicle classification data as proxies, with the SEMIL matrices incorporated in Year 2 once access is formalised. Second, if INFOSIGA-SP geocoding quality at target nodes is insufficient for spatial safety modelling, municipal crash databases at São Carlos and comparison cities will substitute as primary safety data sources, supplemented by INFOSIGA-SP records for severity classification. Third, if DataGEO API integration encounters institutional or technical barriers, SEMIL's published PLI-

SP 2050 planning documents and the publicly accessible DataGEO visualisation portal will provide OD matrix data through manual extraction pipelines documented as part of the replication guidelines. Each contingency has been assessed as compatible with the minimum viable dissertation scope defined in Section 5.7.

6.2 SEMIL DataGEO and Origin-Destination Matrix Deployment

6.2.1 Multimodal OD Matrix Structure and Analytical Application

SEMIL's 62 OD matrices for PLI-SP 2050 characterize freight flows across the state's multimodal network encompassing road, rail, waterway, and pipeline modes for the base planning year. Matrix structure includes spatial dimension (origin-destination zone pairs aligned with PLI-SP 2050 planning units), modal dimension (freight quantities by transport mode and vehicle category), commodity dimension (freight flows by commodity class relevant for infrastructure loading and hazardous materials planning), and temporal dimension (annualized flows with seasonal adjustment factors for major agricultural and industrial freight cycles) (SEMIL, 2026).

The analytical application of these matrices to SIMCU-State's urban node impact assessment requires: identification of OD pairs with significant flows through target urban nodes; temporal disaggregation of annualized flows into hourly arrival profiles at node entry corridors using corridor travel time distributions; spatial allocation of inter-zonal flows to specific urban road links using entropy-based distribution procedures calibrated against observed counts; and iterative calibration of disaggregation parameters using SIMCU-State's real-time urban monitoring data to progressively reduce OD estimation errors.

6.2.2 Complementary State and Municipal Data Sources

Complementary state data sources include: INFOSIGA-SP statewide fatal and injury crash records with geocoding accessible through state research agreement the primary crash dataset for safety

model calibration across urban nodes; DETRAN-SP vehicle registration and classification data providing fleet composition information by municipality for freight exposure estimation; state highway authority (DER-SP) traffic count data for inter-urban corridor loading characterization; and CETESB environmental monitoring data for emissions quantification aligned with PAC 2050 targets.

Municipal data sources retained for local calibration include: traffic camera networks at urban nodes providing real-time vehicle classification and speed data; traffic management center records including signal timing, incident logs, and manual count programs; municipal planning department land use classifications aligned with PLI-SP 2050 socioeconomic characterization zones; and local establishment databases for freight trip generation model calibration. The institutional framework of PLI-SP 2050 provides the intergovernmental coordination mechanism for accessing these municipal datasets through state-municipal data sharing arrangements established in support of SEMIL's planning mandate.

6.3 Supporting State Datasets and Academic Literature

6.3.1 PLI-SP 2050 Supporting Planning Documents

Several PLI-SP 2050 supporting documents provide data and analytical context directly relevant to SIMCU-State's research framework. The Socioeconomic Characterization of São Paulo State provides demographic, economic, and land use data by planning zone that informs freight demand generation modeling and contextualizes urban node safety and congestion outcomes within broader development patterns. The State Climate Adaptation and Resilience Plan (PEARC) establish climate scenario parameters for long-horizon freight infrastructure resilience analysis, connecting SIMCU-State's near-term safety and performance monitoring to PLI-SP 2050's 2050 planning horizon. The State Energy Plan 2050 (PEE 2050) provides decarbonization trajectories

for the state freight sector that translate into vehicle fleet evolution scenarios for long-range safety model projections (SEMIL, 2026).

6.3.2 Academic Literature Aligned with PLI-SP 2050 Research Themes

At the multimodal matrix layer, the geography of warehouses in the São Paulo Metropolitan Region (Guerin et al., 2021) documents the logistics decentralisation process that has created the specific node-level exposure patterns SIMCU-State's OD matrix disaggregation must reproduce. Research on agricultural commodity logistics from a sustainability perspective (Caixeta-Filho et al., 2022) establishes performance evaluation frameworks for São Paulo State's agribusiness freight corridors particularly the soy, sugar, and citrus flows on the SP-310 and SP-330 corridors that SIMCU-State's urban node assessments extend to the safety and congestion dimensions of corridor sustainability. The logistics aspects of transport modalities in raw sugar exports (Vieira et al., 2016) offer multimodal benchmarks for the agribusiness freight that constitutes a significant component of SEMIL's OD matrix flows, providing commodity-specific modal choice evidence relevant to DCEM disaggregation calibration.

At the urban module layer, the sustainability index for integrated urban transport and logistics in São Paulo (Marchetti & Wanke, 2021) provides a composite indicator framework whose components directly overlap with SIMCU-State's monitoring outputs, enabling comparative assessment of urban node performance against the state-wide sustainability benchmarks PLI-SP 2050 is designed to improve. Research on urban logistics and consumer behaviour in São Paulo (Dutra et al., 2021) characterises the demand-side dynamics of last-mile freight e-commerce penetration, delivery frequency, and consumer proximity effects in peripheral versus central zones that PLI-SP 2050's OD matrices capture at the inter-regional scale but require behavioural modelling for urban distribution zone analysis. Research on Logistics 4.0 in Brazil and SDG 9

alignment Silva et al., 2021 frames digitally integrated freight infrastructure as a development imperative that requires the state-to-urban data governance bridges SIMCU-State directly provides.

Collectively, this body of literature traces a clear macro-to-micro analytical path: from the structural inequality geography of São Paulo's freight network (Cardoso, 2025), through the spatial logistics patterns at metropolitan and corridor scale (Guerin et al., 2021; Caixeta-Filho et al., 2022; Vieira et al., 2016), to the urban-level sustainability measurement and demand behaviour frameworks (Marchetti & Wanke, 2021; Dutra et al., 2021) that ground SIMCU-State's outputs in São Paulo State's socioeconomic and multimodal realities. This literature path constitutes the macro analytical foundation upon which SIMCU-State's urban module is built, ensuring that the research's policy recommendations can address not only aggregate efficiency but the equity and sustainability dimensions required by SDG 9 and SDG 11.

7. EXPECTED CONTRIBUTIONS AND SIGNIFICANCE

7.1 Theoretical Contributions

The dissertation delivers several theoretical contributions to the fields of transportation engineering, urban freight management, and state logistics planning, grounded in the macro-to-micro analytical architecture that structures this research:

State-to-urban freight impact translation framework: A validated methodology for disaggregating macro-scale state OD matrices into urban node traffic loading profiles, connecting logistics planning scales that have remained analytically disconnected. This framework addresses the fundamental methodological gap between PLI-SP 2050's state-level planning instruments and the urban-scale impact assessments required for comprehensive externality accounting in state freight policy.

Integrated real-time state-urban monitoring architecture: A unified system design for simultaneous safety and performance monitoring at urban nodes that bridges state logistics planning and municipal traffic management extending the theoretical literature on ITS architecture from city-scale to multi-institutional, multi-scale deployment contexts.

Multimodal OD matrix validation methodology: Techniques for calibrating state-level multimodal freight demand estimates against real-time urban monitoring observations, advancing the methodological foundation for data-quality assessment in large-scale OD surveys a contribution directly applicable to PLI-SP 2050's ongoing matrix refinement processes.

São Paulo State urban node freight dynamics: Empirical characterization of safety and performance patterns at urban nodes across PLI-SP 2050 corridor typologies, filling the research

gap between metropolitan-region studies and the state-wide spatial scope of PLI-SP 2050's planning ambitions.

SDG 9 and SDG 11 alignment framework: A documented methodology for evaluating PLI-SP 2050 freight infrastructure decisions against SDG 9 (resilient, sustainable, and equitable infrastructure) and SDG 11 (safe, inclusive, and sustainable cities) performance criteria, advancing the theoretical literature on how state logistics planning instruments can be designed to produce measurable improvements in the equity and sustainability dimensions of urban freight governance a contribution directly applicable to Brazil's National Transport Plan and to comparable state logistics instruments in other Latin American contexts.

7.2 Practical Contributions

The research provides tangible outputs for PLI-SP 2050 implementation and state transport governance:

Deployable state-urban monitoring bridge: Technical specifications, DataGEO integration protocols, and implementation guidance for extending SIMCU-State's monitoring capabilities to additional urban nodes across PLI-SP 2050's corridor network, with cost estimates scaled to the monitoring investment requirements of different node typologies.

PLI-SP 2050 urban module analytical toolkit: Validated disaggregation procedures, safety and performance prediction models, and scenario analysis capabilities directly compatible with SEMIL's PLI-SP 2050 planning workflows and DataGEO visualization infrastructure.

Evidence base for PLI-SP 2050 intervention evaluation: Quantified relationships between state corridor freight investments, regulatory changes, and urban node safety and congestion outcomes

enabling SEMIL to account for urban externality costs in PLI-SP 2050 project prioritization and policy impact assessment.

Decision support tools for state-municipal integration: Real-time dashboards, alert systems, and scenario analysis capabilities serving both municipal traffic management centers and SEMIL's PLI-SP 2050 planning operations with interface designs explicitly developed through stakeholder engagement with both user communities.

Socioeconomic vulnerability mapping toolkit: A validated methodology for distributional analysis of freight-related safety and congestion impacts across São Paulo's centro-periphery socioeconomic geography, using PLI-SP 2050's Socioeconomic Characterisation zones as the spatial unit of analysis. This toolkit enables SEMIL to identify nodes where freight exposure and socioeconomic vulnerability coincide most acutely, supporting equitable investment prioritisation as required by SDG 11.2's mandate for safe and accessible transport for all. Climate and energy scenario integration: Validated model specifications for projecting how PAC 2050 emissions trajectories, PEARC climate resilience requirements, and PEE 2050 fleet electrification pathways alter safety and performance outcomes at São Paulo State urban nodes over the 2030–2050 horizon, providing SEMIL with the evidence base needed to demonstrate PLI-SP 2050 compliance with São Paulo State's integrated sustainability commitments.

7.3 Policy Implications

The research carries significant implications for state and municipal transport policy across São Paulo:

PLI-SP 2050 urban externality integration: Enabling SEMIL to incorporate urban node safety and congestion costs into PLI-SP 2050's infrastructure investment prioritization transforming the plan's

cost-benefit framework from one that captures corridor-level efficiency gains into one that fully accounts for urban externality consequences.

State-municipal coordination architecture: Providing an operational model for data sharing and joint decision-making between SEMIL and municipal transport agencies a governance contribution that extends beyond SIMCU-State's immediate application to inform the institutional design of future state-municipal transport coordination arrangements.

Equity-centred freight governance: Operationalising Cardoso's (2025) documentation of logistics inequality in the São Paulo Metropolitan Region into actionable policy guidance for PLI-SP 2050 specifically, identifying the peripheral urban nodes where freight corridor investment decisions have the highest absolute safety and congestion impact on socioeconomically vulnerable communities, and providing SEMIL with a distributional impact assessment framework that ensures PLI-SP 2050's investment prioritisation is accountable to SDG 11's inclusive and equitable urban development mandate. This governance contribution extends beyond SIMCU-State's immediate application to inform the design of equity impact assessment requirements in future state freight planning instruments across Brazil.

Sustainable logistics governance, climate resilience, and energy transition: Connecting SIMCU-State's urban impact monitoring to PAC 2050 emissions targets, PEARC climate resilience requirements, and PEE 2050 fleet electrification milestones enabling PLI-SP 2050's sustainability commitments to be tracked and verified through real-time urban node monitoring data. As São Paulo State's commercial vehicle fleet electrifies under PEE 2050, as climate stressors intensify under PEARC's projected scenarios, and as PAC 2050's emissions targets tighten, SIMCU-State's monitoring architecture provides the urban-scale evidence needed to demonstrate that PLI-SP 2050's logistics governance is delivering on its integrated environmental commitments.

Replicable methodology for São Paulo State's urban node inventory: Documenting methodological adaptations for extending SIMCU-State across PLI-SP 2050's full urban node inventory transforming the research from a single-study contribution into a scalable analytical infrastructure for state freight governance.

REFERENCES

- Aquino, É. C., Antunes, J. L. F., & Morais Neto, O. L. (2020). Mortality by road traffic injuries in Brazil (2000–2016): Capital cities versus non-capital cities. *Revista de Saúde Pública*, 54, 122. <https://doi.org/10.11606/s1518-8787.2020054001703>
- Browne, M., Allen, J., & Leonardi, J. (2012). Evaluating the use of an urban consolidation centre and electric vehicles in central London. *IATSS Research*, 35(2), 73–84. <https://doi.org/10.1016/j.iatssr.2011.09.001>
- Caixeta-Filho, J. V., Correa, R. G., & de Lima, J. F. (2022). Performance evaluation of agricultural commodity logistics from a sustainability perspective. *Case Studies on Transport Policy*, 10(2), 1024-1033. <https://doi.org/10.1016/j.cstp.2022.01.029>
- Cardoso, R. (2025). Between rings and radials: Logistics and inequality in the São Paulo Metropolitan Region. *Cadernos Metropolitanos*, 27(52). <https://doi.org/10.1590/2236-9996.2025-6363925-en>
- CNT. (2022). Análise do custo de transporte de cargas no Brasil [Analysis of freight transport costs in Brazil]. Confederação Nacional do Transporte. <https://www.cnt.org.br/estudos/estudo-logistica>
- Dutra, N. G. S., da Silva, J. E. A. R., & Santos, F. A. (2021). Understanding urban logistics and consumer behavior in São Paulo city. *Case Studies on Transport Policy*, 9(3), 1206-1215. <https://doi.org/10.1016/j.cstp.2021.06.017>

Ferreira, M. A., Vieira, J. G. V., Oliveira, R. L. M. de, & Junior, J. C. C. (2023). Medidas de transporte urbano de cargas na Região Metropolitana de São Paulo nas últimas décadas [Urban freight transport measures in the São Paulo Metropolitan Area over recent decades]. *Transportes*, 31(2), e2785. <https://doi.org/10.58922/transportes.v31i2.2785>

Gomes, B. P., Oliveira, L. K., & Bertoncini, B. V. (2022). Modelling freight trip generation based on deliveries for Brazilian municipalities. *Sustainability*, 14(16), 10300. <https://doi.org/10.3390/su141610300>

Gonzalez-Feliu, J., Morana, J., & Salanova, J. M. (2023). Quality of life and urban freight transport: A structural equation modeling approach. *Transportation Research Part D: Transport and Environment*, 118, 103125. <https://doi.org/10.1016/j.trd.2023.103125>

Holguin-Veras, J., Ozbay, K., Kornhauser, A., Brom, M. A., Iyer, S., Yushimito, W. F., & Allen, W. G. (2016). The impacts of off-hour deliveries on urban freight emissions. *Transportation Research Part D: Transport and Environment*, 43, 34–46. <https://doi.org/10.1016/j.trd.2015.11.003>

Institution of Engineering and Technology. (2023). Smart cities: Intelligent transport systems and urban mobility. IET Standards.

Isler, C. A., Huang, Y., & de Melo, L. E. A. (2024). Developing accident frequency prediction models for urban roads: A case study in São Paulo, Brazil. *IATSS Research*, 48(2). <https://doi.org/10.1016/j.iatssr.2024.03.003>

Lord, D., & Mannering, F. (2010). The statistical analysis of crash-frequency data: A review and assessment of methodological alternatives. *Transportation Research Part A: Policy and Practice*, 44(5), 291-305. <https://doi.org/10.1016/j.tra.2010.02.001>

Machado, S. T., & dos Reis, J. G. M. (2022). Fatal traffic accidents in São Paulo State, Brazil: Social characteristics and social distance effects. In *IFIP Advances in Information and Communication Technology* (Vol. 663, pp. 574–581). Springer. https://doi.org/10.1007/978-3-031-16407-1_67

Marchetti, D., & Wanke, P. (2021). An index for the sustainability of integrated urban transport and logistics: The case study of São Paulo. *Sustainability*, 13(21), 12116. <https://doi.org/10.3390/su132112116>

McDonald, N. C., Zhang, Y., & Schimek, P. (2019). Trends in fatal crashes involving large trucks and passenger vehicles, 2009–2015. *Traffic Injury Prevention*, 20(8), 823–828. <https://doi.org/10.1080/15389588.2019.1630531>

Oliveira, G. F., de Oliveira, L. K., & de Lima, S. M. (2017). Environmental impacts of urban freight transport: A life cycle assessment approach. *Transportation Research Part D: Transport and Environment*, 55, 131–145. <https://doi.org/10.1016/j.trd.2017.09.014>

Guerin, L., Vieira, J. G. V., de Oliveira, R. L. M., de Oliveira, L. K., Vieira, H. E. M., & Dablan, L. (2021). The geography of warehouses in the São Paulo Metropolitan Region and contributing

factors to this spatial distribution. *Journal of Transport Geography*, 91, 102976.
<https://doi.org/10.1016/j.jtrangeo.2021.102976>

Setti, J. R. (2009). Highway capacity and level of service: Brazilian perspectives. *Transportation Research Record*, 2130, 89–97. <https://doi.org/10.3141/2130-11>

SEMIL. (2026). Plano Logístico e de Infraestrutura do Estado de São Paulo PLI-SP 2050 [São Paulo State Logistics and Infrastructure Plan PLI-SP 2050]. Secretaria de Meio Ambiente, Infraestrutura e Logística. <https://pli.semil.sp.gov.br/>

△ Silva, [First Initial]. [Additional Authors]. (2021). [Logistics 4.0 in Brazil and alignment with SDG 9: Full title to be confirmed]. [Journal/Proceedings to be confirmed]. [Volume/Issue to be confirmed]. [DOI to be confirmed]. Note to researcher: This reference was provided by Prof. André as ‘Logistics 4.0 in Brazil and SDG 9 (2021)’. Full bibliographic details including author names, journal, volume, and DOI require verification and completion before submission. Do not cite in final document until fully verified.

Soimun, F., Sudira, P., & Dharma, A. (2025). Analysis of freight transport efficiency and congestion impacts in the Denpasar-Gilimanuk corridor. *Transportation Research Procedia*, 82, 274-281.

Vieira, J. G. V., Fransoo, J. C., & Carvalho, C. D. (2016). Logistics aspects of transport modalities on the exports of raw sugar. *European Scientific Journal*, 12(16), 345.
<https://doi.org/10.19044/esj.2016.v12n16p345>

Wang, H., & Holguin-Veras, J. (2008). Tour-based entropy maximization formulation of urban freight demand. *Transportation Research Record*, 2066, 47–54. <https://doi.org/10.3141/2066-06>

World Bank. (2023). Analysis of shared space in the city of São Paulo. World Bank Group. <https://documents.worldbank.org/en/publication/documents-reports>

European Commission. (2022). FENIX: Federated European Network of Information eXchange in logistics Project summary and results. Directorate-General for Mobility and Transport. <https://transport.ec.europa.eu/transport-themes/intelligent-transport-systems/road>

Cidell, J. (2011). Flows and interpretations: The new economic geography of freight logistics. *Urban Geography*, 32(7), 1047–1065. <https://doi.org/10.2747/0272-3638.32.7.1047>

Taniguchi, E., Thompson, R. G., & Yamada, T. (2014). New opportunities and challenges for city logistics. *Transportation Research Procedia*, 2, 1–11. <https://doi.org/10.1016/j.trpro.2014.09.001>

del Esposte, A., Santana, E., Kanashiro, L., Cerqueira, F., Bizarro, P., & Braghetto, K. R. (2017). InterSCity: A scalable microservice-based open source platform for smart cities. *Proceedings of the 6th International Conference on Smart Cities and Green ICT Systems*, 35–46. <https://doi.org/10.5220/0006306600350046>

Hauer, E., Harwood, D. W., Council, F. M., & Griffith, M. S. (2002). Estimating safety by the Empirical Bayes method: A tutorial. *Transportation Research Record*, 1784, 126–131. <https://doi.org/10.3141/1784-16>

IBGE. (2022). Censo demográfico 2022: Resultados do universo [2022 Demographic Census: Population results]. Instituto Brasileiro de Geografia e Estatística. <https://censo2022.ibge.gov.br/>

IBGE. (2023). Produto interno bruto dos municípios [Gross domestic product of municipalities]. Instituto Brasileiro de Geografia e Estatística. <https://www.ibge.gov.br/estatisticas/economicas/contas-nacionais/9088-produto-interno-bruto-dos-municipios.html>

IEEE. (2019). IEEE Std 2413-2019: IEEE standard for an architectural framework for the Internet of Things (IoT). Institute of Electrical and Electronics Engineers. <https://doi.org/10.1109/IEEESTD.2020.9023007>

ISO. (2020). ISO 21217:2020 Intelligent transport systems Communications access for land mobiles (CALM) Architecture. International Organization for Standardization. <https://www.iso.org/standard/74430.html>